

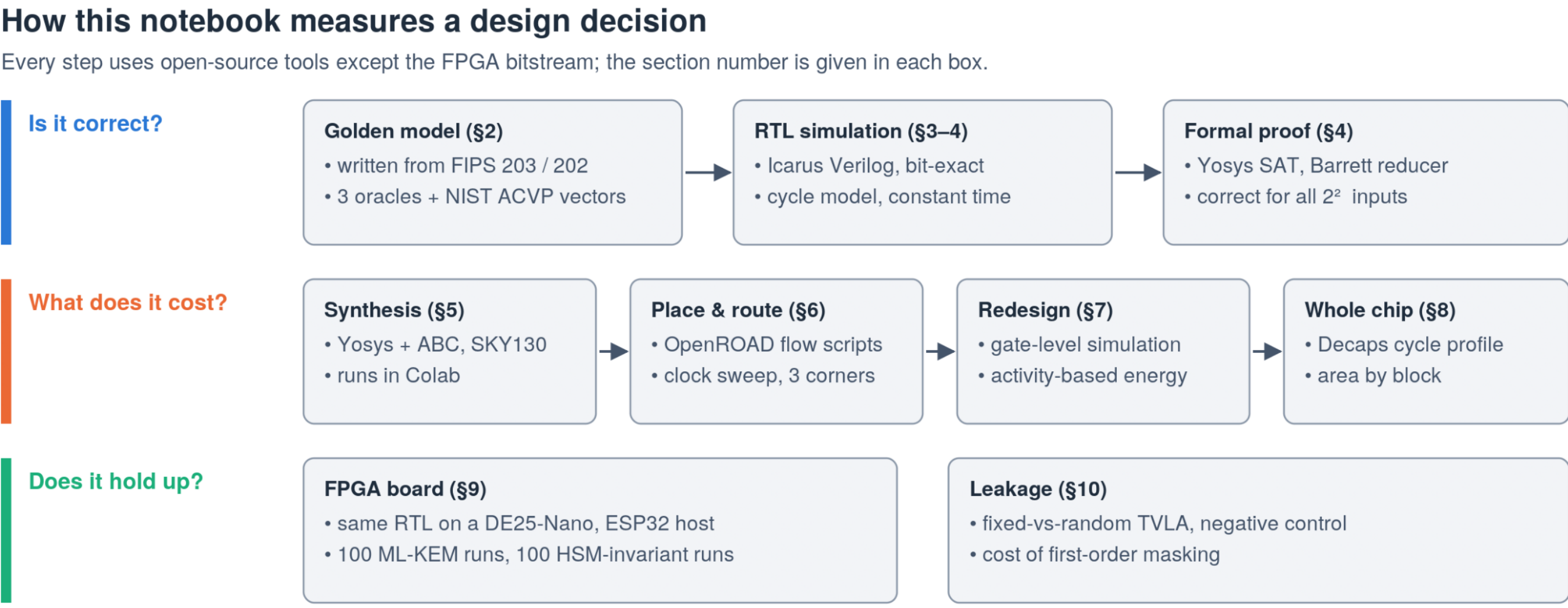
Measuring the Design Decisions of an Open-Source Post-Quantum HSM Chip

ML-KEM-512 NTT and Keccak, from a Python golden model to a SKY130 layout

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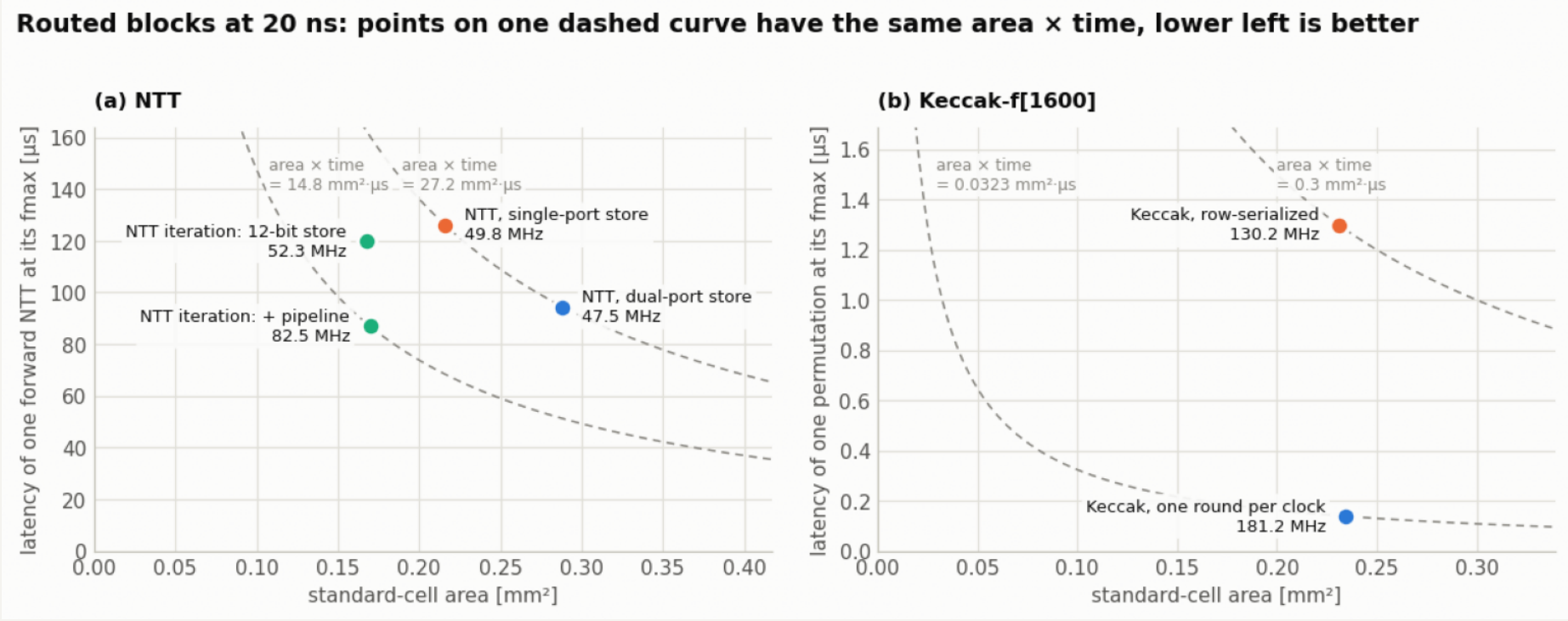
What did each architectural decision made for the ASIC actually cost in area, latency and energy, what does the design still leak, and can anyone re-derive those numbers with open tools?



I. Is it correct?

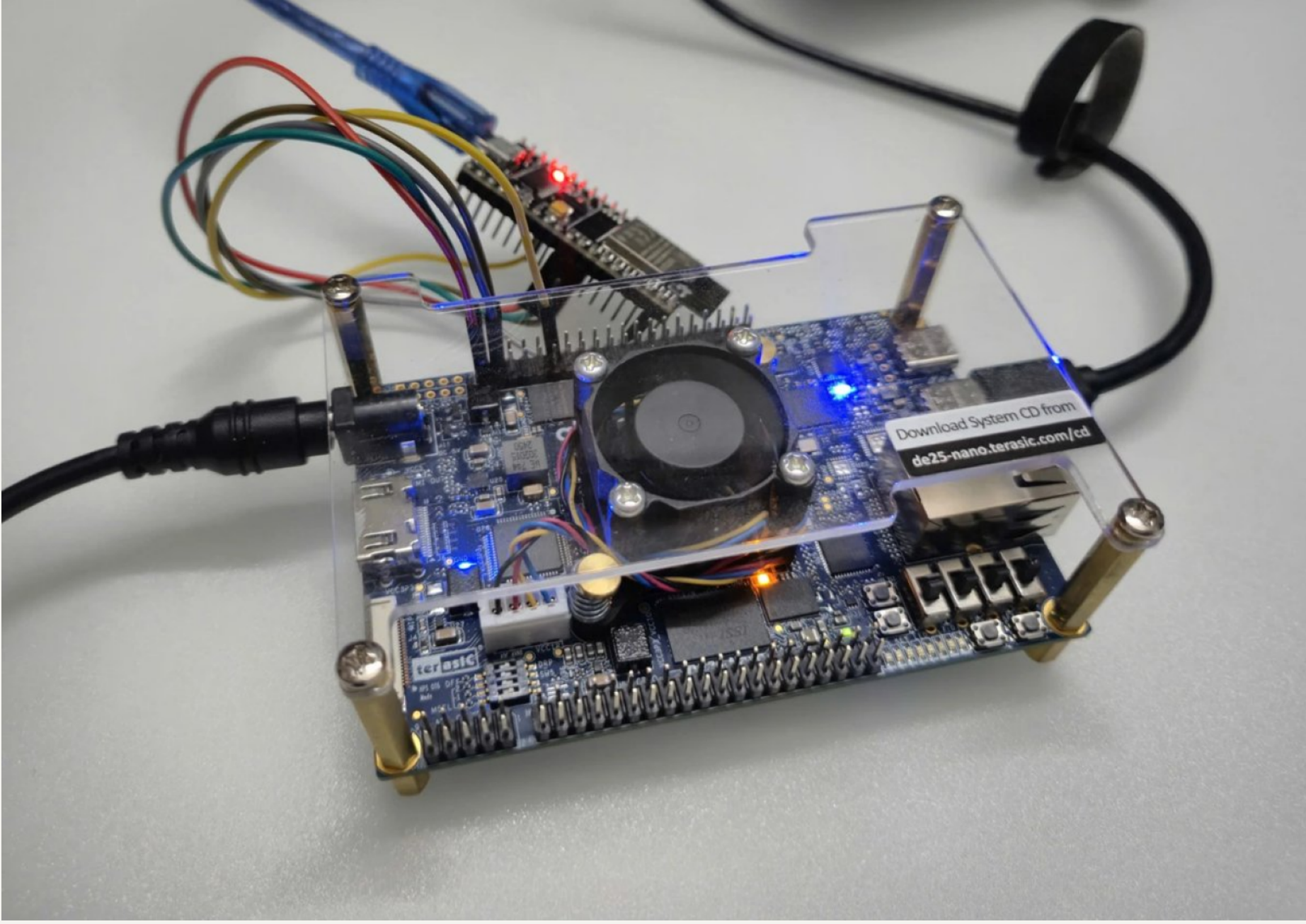
- 25/25 NIST ACVP key generations reproduced byte for byte by the complete RTL
- 0 mismatches over 208,896 NTT coefficients and 404 Keccak permutations, at a constant latency
- The Barrett reducer is proven correct for all 2^{24} inputs, and one subtraction suffices
- The routed netlists reproduce the golden results at gate level

II. What does it cost?

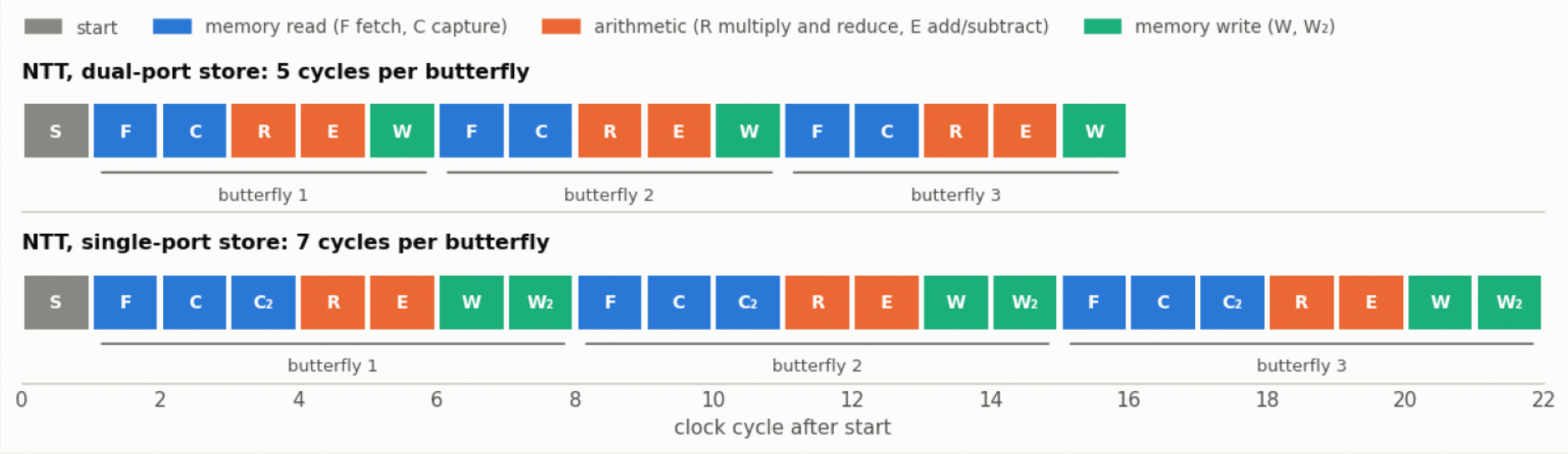


A measured redesign of the NTT, after routing: fmax 50 → 83 MHz, area -21%, area × time -46%.

III. Does it hold up?



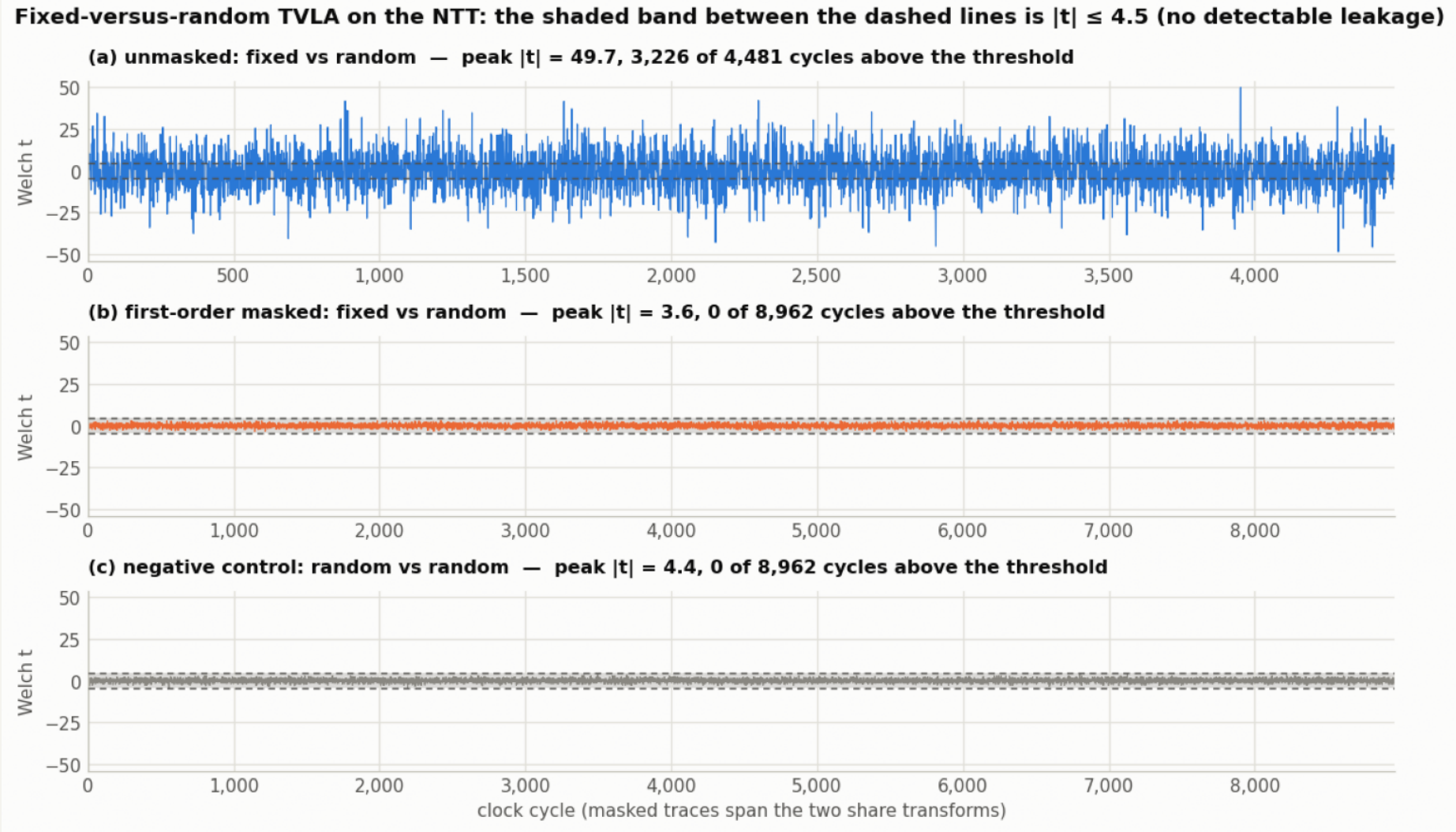
The same RTL on the DE25-Nano: 100/100 two-role ML-KEM runs pass.



The controller visits the 896 butterflies in the order of FIPS 203, Algorithm 9; the single-port store needs two extra cycles per butterfly.

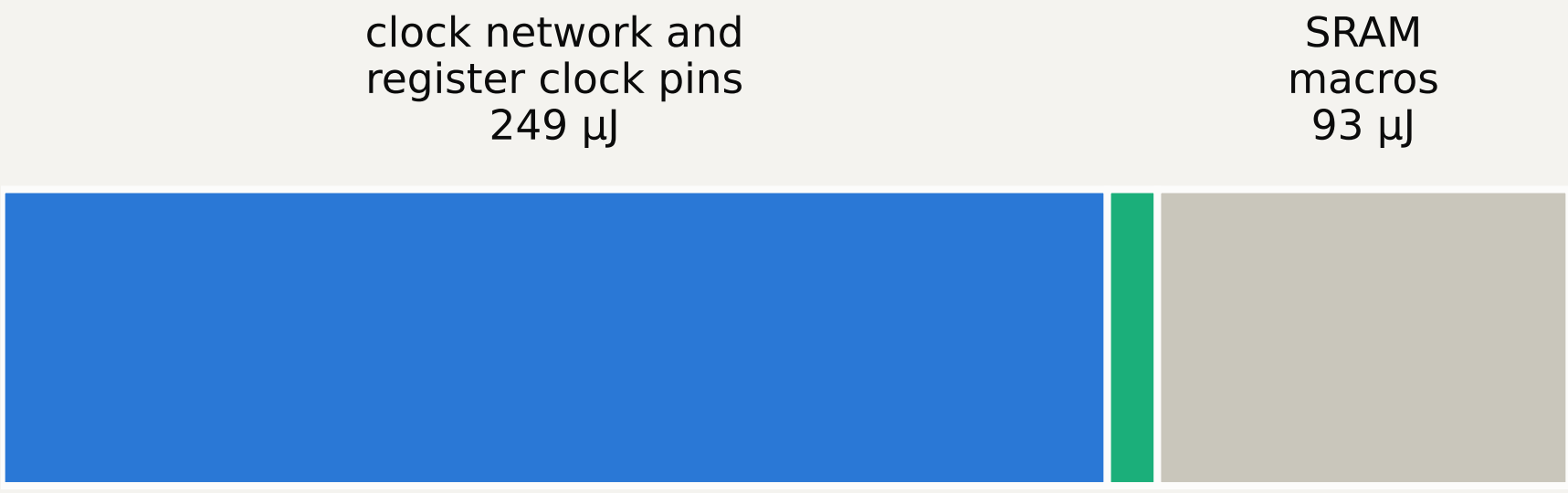
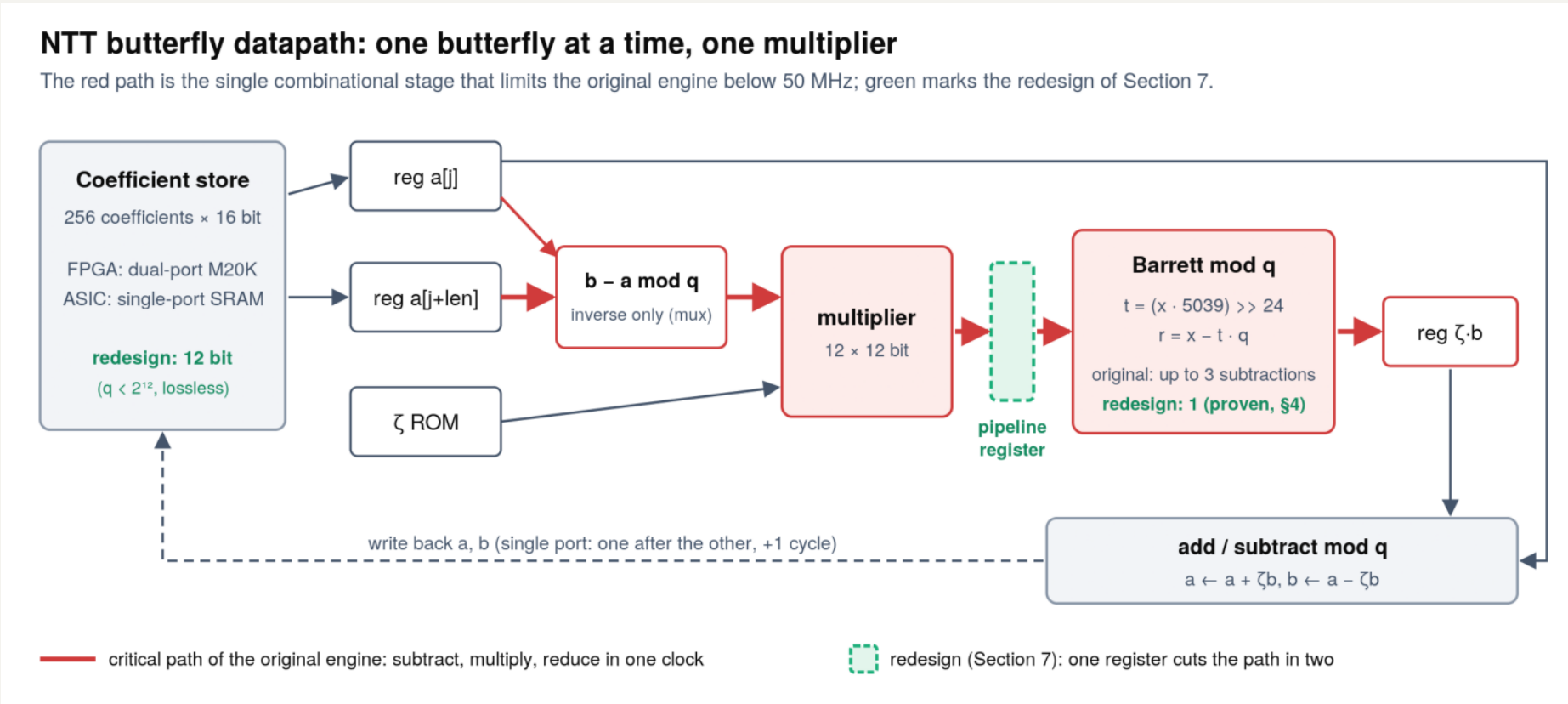


Energy per forward NTT: 2.48 µJ with a flip-flop store, 0.50 µJ with the chip's SRAM macro.



Simulated TVLA: largest |t| 49.7 unmasked, 3.6 with first-order masking (threshold 4.5).

One decapsulation on the chip: 0.35 mJ at 25 MHz



The integration, not ML-KEM, sets the energy: gating the clock of idle blocks and the chip selects of the macros would save roughly 40 to 55 %. The NTT is busy 50% of a decapsulation, the Keccak permutation 0.8%.

What transfers

- Weigh a block by how often the system waits for it.
- Measure the integrated chip, not only its blocks.
- Check the check: six results here first looked right and were not.
- Tie the prose to the data: assertions stop the notebook when they part.